

$$A_P = {}^A R_B \cdot B_P$$

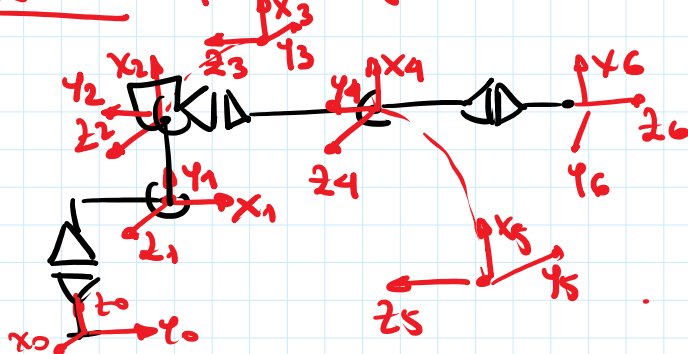
$$R = (\cos \theta, \sin \theta)$$

$$K = (3, 2, 1) = u$$

$$\hat{u} = \frac{3, 2, 1}{\|u\|}$$

$$r = \begin{bmatrix} 5 \\ 2 \\ 6 \end{bmatrix}$$

VRL-2015 (Ejercicio de Gpde).



$$\| \text{rot}_f(\theta + \phi) \| = \| \text{rot}_f(\theta) \| \cdot \| \text{rot}_f(\phi) \|$$

Cinemática Diferencial

apal (Sara TS 2015).

$$\dot{p} = J \dot{q} = \frac{d}{dt} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$x = L_1 C_1 + L_2 C_{12}$$

$$y = L_1 S_1 + L_2 S_{12}$$

$$z = d_3$$

$$u = \theta_1 + \theta_2 + \theta_4$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \frac{d}{dt} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{d}{dt} (L_1 C_1 + L_2 C_{12}) \\ \frac{d}{dt} (L_1 S_1 + L_2 S_{12}) \\ \frac{d}{dt} (d_3) \end{bmatrix} = \begin{bmatrix} -L_1 S_1 \dot{\theta}_1 - L_2 S_{12} (\dot{\theta}_1 + \dot{\theta}_2) \\ L_1 C_1 \dot{\theta}_1 + L_2 C_{12} (\dot{\theta}_1 + \dot{\theta}_2) \\ -\dot{d}_3 \end{bmatrix}$$

$$\therefore \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} (-L_1 S_1 - L_2 S_{12}) \dot{\theta}_1 - L_2 S_{12} \dot{\theta}_2 \\ (L_1 C_1 + L_2 C_{12}) \dot{\theta}_1 + L_2 C_{12} \dot{\theta}_2 \\ \dot{d}_3 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} -L_1 S_1 - L_2 S_{12} & -L_2 S_{12} & 0 \\ L_1 C_1 + L_2 C_{12} & L_2 C_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{d}_3 \end{bmatrix}$$

$$\begin{aligned} x &= L_1 C_1 + L_2 C_{12} \\ y &= L_1 S_1 + L_2 S_{12} \\ z &= d_3 \end{aligned}$$

Velocidades lineales

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} & \frac{\partial x}{\partial d_3} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} & \frac{\partial y}{\partial d_3} \\ \frac{\partial z}{\partial \theta_1} & \frac{\partial z}{\partial \theta_2} & \frac{\partial z}{\partial d_3} \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{d}_3 \end{bmatrix}$$

JJ = simplify(J)

JJ =

$$\begin{bmatrix} -L_2 \sin(\theta_1 + \theta_2) - L_1 \sin(\theta_1) & -L_2 \sin(\theta_1 + \theta_2) & 0 \\ L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) & L_2 \cos(\theta_1 + \theta_2) & 0 \end{bmatrix}$$

$$J = \begin{pmatrix} -L_2 \sin(\theta_1 + \theta_2) - L_1 \sin(\theta_1) & -L_2 \sin(\theta_1 + \theta_2) & 0 \\ L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) & L_2 \cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\dot{\theta}_1 \quad \dot{\theta}_2 \quad \dot{\theta}_3 \quad \dots$$

Jacobiano, Lineal.

Jacobiano Angular

$${}^i \omega_i = \underline{\dot{q}_i} K; K = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$${}^0 \bar{\omega}_n = {}^0 \bar{\omega}_1 + {}^0 \bar{\omega}_2 + {}^0 \bar{\omega}_3 + \dots + {}^0 \bar{\omega}_n$$

$${}^0 \bar{\omega}_n = {}^0 \bar{\omega}_1 + {}^0 R_1 {}^1 \omega_2 + {}^0 R_2 {}^2 \omega_3 + \dots + {}^0 R_{n-1} {}^{n-1} \omega_n$$

$${}^0 \bar{\omega}_n = \dot{q}_1 K + {}^0 R_1 \dot{q}_2 K + {}^0 R_2 \dot{q}_3 K + \dots + {}^0 R_{n-1} \dot{q}_n K$$

$${}^0 \bar{\omega}_n = \dot{q}_1 z_0 + \dot{q}_2 z_1 + \dot{q}_3 z_2 + \dots + \dot{q}_n z_{n-1}$$

$${}^0 \bar{\omega}_n = \underbrace{[\dot{q}_1 z_0 \quad \dot{q}_2 z_1 \quad \dot{q}_3 z_2 \quad \dots \quad \dot{q}_n z_{n-1}]}_{\text{Jacobiano angular}} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \vdots \\ \dot{q}_n \end{bmatrix}; e_i = \begin{cases} 1 & \text{Art. Rot} \\ 0 & \text{Art. Primi} \end{cases}$$